Due: Thursday April 14th, latest at 2PM

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- This problem set is about calculating electric fields in electrostatic problems and magnetic fields in magnetostatic problems (using the principle of superposition), and approximating these expressions at large distances. You should be learning about electric dipoles and magnetic dipoles from this problem set.
- Wherever appropriate, do remember to provide a sketch.
- You might need to use the binomial approximation: $(1+x)^\alpha \approx 1+\alpha x$ when $|x| \ll 1$. For example, if $D \ll C$, then

$$\frac{1}{(C+D)^{3/2}} = \frac{1}{C^{3/2}}\left(1+\frac{D}{C}\right)^{-3/2} \approx \frac{1}{C^{3/2}}\left(1-\frac{3}{2}\frac{D}{C}\right)$$

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1. Two point charges with charge $+q$ and $-q$ lie on the y-axis, respectively at $(0, d/2, 0)$ and $(0, -d/2.0)$. This is the simplest example of an electric dipole, with dipole moment $p = qd$.

   (a) **[1 pt]** Consider the electric field at the point $(x_0, 0, 0)$ on the x-axis. Write down the result for the electric field.

   (This was calculated in class, in tutorial, and in problem set 01, so there's no need to show the derivation. However, if the calculation is not familiar to you, it would be an excellent idea to re-learn it now.)

   (b) **[3 pts]** Approximate the electric field for $x_0 \gg d$, expressing your result in terms of the electric dipole moment p.

   (c) **[1 pt]** The electric field due to a single charge (monopole) decreases as the inverse square of the distance. Describe (in one sentence or phrase) the distance-dependence of the electric field due to an electric dipole, at large distances.

   (d) **[3 pts]** Consider the point $(0, y_0, 0)$ on the y-axis, with $y_0 > d$. Derive an expression for the electric field at this point.

(e) [**4+1 pts**] Approximate this expression for $y_0 \gg d$. Describe the distance-dependence in this direction.

2. A thin glass rod of length L lies along the x axis with one end at the origin and the other end at $(L, 0, 0)$. The rod carries a uniformly distributed postive charge Q.

   (a) [**6 pts**] Consider the point $(d, 0, 0)$, also on the x axis, with $d > L$. Calculate the electric field generated at this point. You will have to first consider an infinitesimal element of the rod, and then integrate over appropriate limits.

   (b) [**4+1 pts**] Consider the limit $d \gg L$ and approximate your result for this limit. Explain why you could have expected this result for the electric field far from the rod.

3. Two thin rods, each of length L, are placed along the y-axis, with one end of each at the origin. The rod on the positive y axis, with other end at $(0, L, 0)$, carries a uniformly distributed postive charge Q. The rod on the negative y axis, with other end at $(0, -L, 0)$, carries a uniformly distributed negative charge $-Q$. This is a more complicated electric dipole.

   (a) [**6 pts**] Consider the point $(0, y_0, 0)$ on the y-axis, with $y_0 > L$. Use the result of problem 2a to derive an expression for the net electric field at this point.

   (b) [**4 pts**] For $y_0 \gg L$, approximate this expression to obtain a field of the form

   $$E \approx \frac{1}{4\pi\epsilon_0} \frac{2p}{y_0^3}$$

   Identify p in terms of Q and L. This quantity serves as the electric dipole moment for the present charge configuration.

4. A ring of wire with radius R is centered at the origin and lies on the x-y plane, so that it's axis coincides with the z-axis. Current I flows around this circular wire loop.

(a) **[1+3 pts]** We considered in class the magnetic field created on the axis, away from the plane of the ring.

Write down (from class or textbook) the expression for the magnetic field at the point $(0, 0, z_0)$. Derivation is not necessary.

Approximate this expression for the limit $z_0 \gg R$, and show that the field at large distances has the form

$$B \;\approx\; \frac{\mu_o}{4\pi}\frac{2m}{z_0^3}$$

Express m in terms of the current and the area of the ring. This quantity is the magnetic dipole moment. (A loop of current functions as a magnetic dipole.)

(b) **[6 pts]** Consider a point on the plane of the ring (x-y plane), at a distance x_0 from the center. Let's take this point to be on the x-axis. Using the Biot-Savart law, it can be shown that the magnetic field produced by the ring at this point (for both $x_0 < R$ and $x_0 > R$) has magnitude

$$B = 2\frac{\mu_0 I}{4\pi}\int_0^{\pi} \frac{(R - x_0\cos\phi)R d\phi}{(R^2 + x_0^2 - 2x_0 R\cos\phi)^{3/2}}$$

(You might want to derive this for yourself by drawing careful detailed scketches including all relevant lengths and angles, but please don't submit the derivation. Also, don't bother trying to perform this integral; it is not expressible in terms of common functions.)

Consider the limit $x_0 \gg R$. In this limit, we can approximate the integrand so that the integral can be performed. First, neglect the R^2 term in the denominator, which is permissible because $R^2/x_0^2 \ll R/x_0 \ll 1$. Then use the binomial approximation, and again neglect any term with R^2/x_0^2. Performing the integral, show that the field at large distances has the form

$$B \;\approx\; \frac{\mu_o}{4\pi}\frac{m}{x_0^3}$$

5. An electric dipole with dipole moment $\mathbf{p}$ at the orign is known to generate the following electric field at point $\mathbf{r}$, provided that $r = |\mathbf{r}|$ is large:

$$\mathbf{E} \;=\; \frac{1}{4\pi\epsilon_0}\frac{3(\mathbf{p}\cdot\hat{\mathbf{r}})\hat{\mathbf{r}} - \mathbf{p}}{r^3} \;=\; \frac{1}{4\pi\epsilon_0}\left(\frac{3(\mathbf{p}\cdot\mathbf{r})\mathbf{r}}{r^5} - \frac{\mathbf{p}}{r^3}\right)$$

where $\hat{\mathbf{r}}$ is a unit vector in the direction of $\mathbf{r}$.

(a) [**4 pts**] The cases we have considered in this problem set involve $\mathbf{r} \perp \mathbf{p}$ and $\mathbf{r} \parallel \mathbf{p}$. Simplify the above expression for these two special cases.

(It's worth pondering over whether your results now match your results from problems 1 and 3.)

(b) [**2 pts**] You should be able to guess a very similar expression for the magnetic field far from a magnetic dipole with magnetic dipole moment $\mathbf{m}$. Write down this expression. (It's okay to check your answer from a textbook or website.)